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New Ankle Actuation Mechanism for a Humanoid Robot ⋆

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Abstract: In this article we discuss the design of a new ankle actuation mechanism for the humanoid robot TUlip. The new mechanism consists of two coupled series-elastic systems. We discuss the choice of actuators according to calculations for maximum achievable walking speed. Some control issues, MIMO and non-linearities are also discussed.

Keywords: Walking, Mechanisms, Actuators; Humanoid Robots, Series elastic actuation.

1. INTRODUCTION

The field of humanoid robots gets more and more attention nowadays. As batteries, actuators and computers have become better, it has been increasingly more easy to build nice-looking well-working robots from off-the-shelf components.

In a collaboration project, the three Technical Universities of the Netherlands (University of Twente, University of Delft and University of Eindhoven) are developing a humanoid robot called TUlip (Fig. 1). Contrary to most other humanoid robot projects, where the focus lies on developing advanced controllers for a mechanical system based on very simple principles [1] , we try to follow the more mechatronic approach: solve each problem in the domain where it can be solved best. As an example, the hips and knees of TUlip are actuated by a *series elastic actuator* (SEA) (Williamson, 1995). Amongst others, this makes the system less vulnerable for shocks, for example on foot impact. Hence, at the control we don't have to cope with avoiding shocks anymore.

The main research goal for TUlip is research on 'limit cycle walking' (Hobbelen, 2008). In limit cycle walking, the goal is not to be stable at each time instant, but just have a stable cycle; during the cycle the robot can be unstable, *i.e.*, 'falling' for some time. This is in contrast to the so-called 'ZMP'-type of walking, where tight control makes sure that the robot is always fully supported by the floor.

TUlip is 1.37 m in height and has a mass of 19 kg. It has 17 degrees of freedom: six in each leg, one in each arm and three in the neck. The fore-aft motion of the hip and the knee are equipped with a series elastic actuator.

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[1] Although these robots are, from a mechanical point of view, amazingly nicely designed, the underlying principle is old-fashioned: just heavily geared motors on each of the joints. This is in contrast to humans, which have a great level of compliancy in their muscles.



Fig. 1. TUlip.

The original ankle mechanism on the robot had one active (fore-aft; 'ankle-y') and one passive (left-right; 'ankle-x') degree of freedom and did not allow the robot to balance on one leg (it would fall sideways). It also appeared that the actuation principle for ankle-y was far from satisfactory. In order to improve the performance of TUlip, a new ankle mechanism was designed and built, which is the topic of this paper.

This paper is organized as follows. In section 2 the previous design of the ankle is described. We show what was wrong with it and how it could be solved. Then, in section 3 we will investigate in more detail what the new ankle actuation mechanism should be capable of. The design is discussed in 4. Before concluding we dive into control issues of the system in section 5.

2. OLD ANKLE DESIGN

A schematic view of the old ankle design is shown in Fig. 2. There are two rotation axes involved: 'ankle-x' (for sideways movement) and 'ankle-y' (for fore-aft movement). As shown in the figure, these are not coupled. The two joints are discussed one by one here.

2.1 Sideways joint (ankle-x)

There is no actuator for sideways movement; instead the foot is kept more or less in its default position by two springs ($K \approx 15$ kN/m). The springs are not strong enough to stabilize the robot passively when standing on one leg; neither were they intended to. The idea behind this

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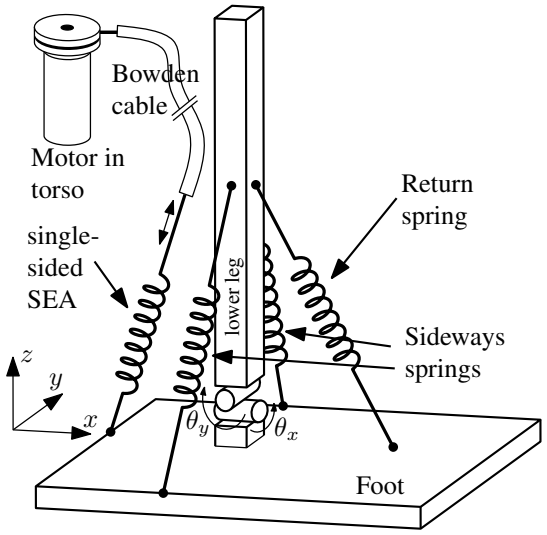


Fig. 2. Schematic view of the old ankle mechanism.

set-up was to allow the robot to rock from one foot to the other (instead of going to a stable equilibrium on one leg); a prerequisite for dynamic walking. The rocking frequency can be tuned by changing the equilibrium position of the springs.

However, standing on one foot during a kick is really hard. Using Acrobot-like control (Hauser and Murray, 1990) was considered nearly impossible, due to the limited actuation range in the hip and the need to do other tasks simultaneously, such as ball kicking. Moreover, severe torso motions make it impractical or even dangerous to catch the robot when it is falling. Therefore, we decided not to try this and improve the mechanics instead.

2.2 Fore-aft joint (ankle-y)

The fore-aft motion is unilaterally actuated, *i.e.*, the foot can be rotated actively in one direction but not in the other. A DC motor placed in the torso is connected to a bowden cable with series-elastic spring, which in turn is connected to the rear of the foot. Pulling the bowden cable makes the foot plantarflex (*i.e.*, toes down). A return spring on the front of the foot makes the foot dorsiflex (*i.e.*, toes up) again when the cable is loosened again.

The main reason why this design choice was made was to keep the weight of the lower leg and foot as low as possible. By putting the motor in the torso, the lower leg and foot just had to be some structural pieces of aluminum, without any heavy structures for actuation. The system however did not work as expected. In order to be able to dorsiflex the foot with a large enough torque, the return spring had to be quite stiff. This implied that the actuator had to deliver a large torque just to keep the foot still (in order to obtain a zero resultant torque). The induced tension in the bowden cable in its turn caused an enormous amount of friction, which made it virtually impossible to do precise control.

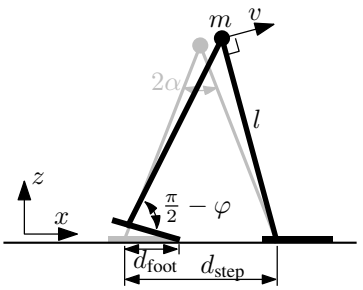


Fig. 3. Simple 2D model of our robot: a compass walker with feet.

3. REQUIREMENTS FOR NEW DESIGN

For the new actuator mechanism we had a few qualitative as well as quantitative requirements. These are described below.

Firstly, the system should not suffer from the problems described in the previous section, so unilateral actuation, bowden cables and under-actuation were not an option. Furthermore, the system should be robust and as lightweight as possible. If mounted on the lower leg, its position should be as high as possible in order not to disturb the passive-dynamic swing properties of the leg too much (Franken, 2007). One of the demands is that the actuation is done in a series-elastic way. This reduces shock loads on the motors and gives us the possibility to mimic zero-torque control.

Quantitatively, the most demanding task for the system is supplying enough energy during push-off in order to compensate for the energy lost during impact. This is not only a matter of much torque, but also of speed: the foot must plantarflex fast enough in order stay in contact with the ground during push-off.

For the determination of these requirements, we look at the system as if it were a 'simplest walker' (Garcia et al., 1998) with flat feet: a 2D, bipedal walker with a point mass m at the hip and negligible foot mass (Fig. 3). We assume post-impact push-off, which is the first strategy that we want to implement using the new ankle mechanism [2]. The model used implies an instantaneous inelastic swing foot collision. Contrary to the simplest walker without feet, the double support phase of this model is not instantaneous: the toe of the new swing foot keeps in touch with the ground for a short period t_p. During this period, the ankle actuator can exert a torque which results in an acceleration of the hip.

A few important parameters of the actual robot (that are used in the compass model as well) are listed in table 1.

3.1 Acceleration dependency

Denote the velocity of the hip by v and its acceleration by a. Assume that just before foot impact the velocity of the walker is v_{pre}. Because of impact loss, the velocity just

[2] Pre-impact push-off has been shown to be more energy efficient than post-impact push-off (Kuo, 2002), so if the new mechanism can achieve post-impact push-off, it is also able to do pre-impact push-off.

Parameter	Name	Value
Leg length	l	0.55 m
Step length	d_{step}	0.40 m
Mass	m	19 kg
Foot length	d_{foot}	0.145 m
Foot rotation limit	$-0.37 < \varphi < 0.18$	rad
Foot rotation stroke	$\Delta\varphi$	0.55 rad

Table 1. Robot parameters for the compass walker model.

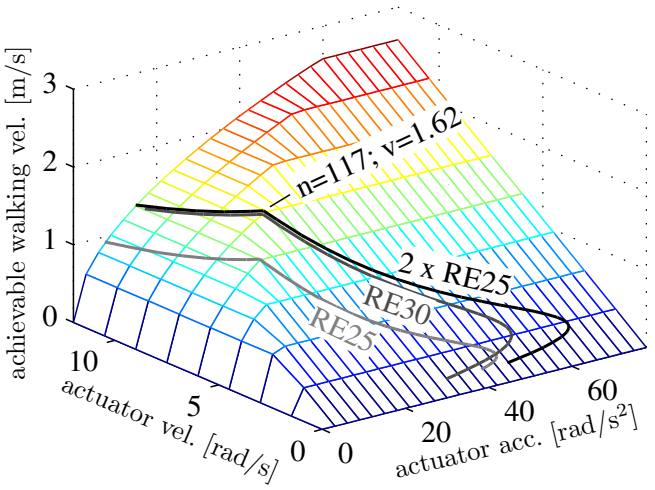


Fig. 4. The maximum achievable walking velocity as a function of the maximum actuator angular acceleration and angular velocity, as described in section 3. The three thick lines correspond to a single Maxon RE25 motor, two RE25's (differential setup) and a single Maxon RE30 motor, with varying gear reduction ratios (section 4.3). The best walking velocity ($v = 1.62$ m/s) is obtained for the differential setup of two RE25 motors with (twice) a gearbox reduction of 1:117.

after impact (but before push-off) is $v_{\text{post}} = v_{\text{pre}} \cos(2\alpha)$. So the kinetic energy loss is

$$E_{\text{impact-loss}} = \tfrac{1}{2}m(v_{\text{pre}}^2 - v_{\text{post}}^2) = \tfrac{1}{2}m(1 - \cos^2(2\alpha))\, v_{\text{pre}}^2. \tag{1}$$

The injected energy by the actuator is the exerted torque multiplied by the stroke, *i.e.*,

$$E_{\text{injected}} = \tau \cdot \Delta\varphi = J\ddot{\varphi} \cdot \Delta\varphi \tag{2}$$

where J is the apparent inertia moment, *i.e.*, the amount of torque required for one unit rotational acceleration, which can be calculated as follows. The relation between hip velocity v and foot rotation velocity $\dot{\varphi}$ and between acceleration a and $\ddot{\varphi}$ are approximately

$$v \approx -\frac{d_{\text{foot}}}{\sin 2\alpha}\dot{\varphi} \Rightarrow a \approx -\frac{d_{\text{foot}}}{\sin 2\alpha}\ddot{\varphi}. \tag{3}$$

Because we assume the only mass is in the hip, all power delivered by the ankle actuator will flow into acceleration of the mass, *i.e.*,

$$P = \tau\,\dot{\varphi} = F_v\, v \Rightarrow J\ddot{\varphi}\,\dot{\varphi} = ma\, v \tag{4}$$

where F_v is the component along the velocity vector v of the resulting force on the hip. Filling in (3) yields

$$J \approx \left(\frac{d_{\text{foot}}}{\sin 2\alpha}\right)^2 m. \tag{5}$$

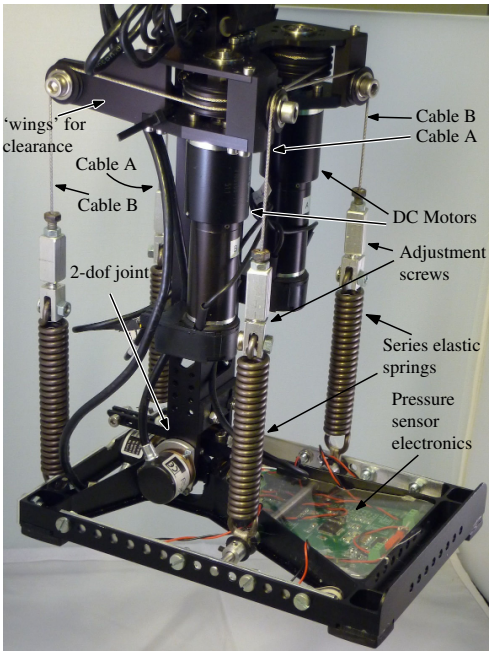


Fig. 5. TUlip's new ankle mechanism.

In order to obtain a limit cycle, it is necessary that the kinetic energy of each step is the same. This implies that the amount of injected energy should equal the amount of lost energy, *i.e.*, $E_{\text{impact-loss}} = E_{\text{injected}}$. Combining (1) and (2), we can calculate the achievable walking velocity v for any acceleration $\ddot{\varphi}$:

$$v_{\text{pre}} = \sqrt{\frac{2J\Delta\varphi}{m(1 - \cos^2(2\alpha))}\ddot{\varphi}}. \tag{6}$$

3.2 Velocity dependency

At the end of the push-off phase, the foot must rotate fast in order to keep the toe on the floor (otherwise, it is not able anymore to push-off). Equation (3) tells us that, when we have a certain maximum rotational velocity $\dot{\varphi}$, the walker can walk at a certain maximum velocity, being

$$v \approx -\frac{d_{\text{foot}}}{\sin 2\alpha}\dot{\varphi}. \tag{7}$$

In order to walk at a desired velocity v, we need both a minimum acceleration (equation (6)) and a minimum velocity (equation (7)), *i.e.*,

$$v_{\text{achievable}} = \min(v_{\text{achievable,acc}}, v_{\text{achievable,vel}}). \tag{8}$$

In Fig. 4 the maximum achievable velocity for our walker is shown as a function of the maximum acceleration and velocity that an actuator can generate. Now, for any choice of actuator, we can calculate the maximum velocity and acceleration and, from the graph, read out the maximum achievable walking velocity. This will be done in section 4.3.

4. NEW DESIGN

In this section we describe the innovative design of the new ankle actuation mechanism. Firstly we give an overview of the system, then we discuss some of the properties in more detail.

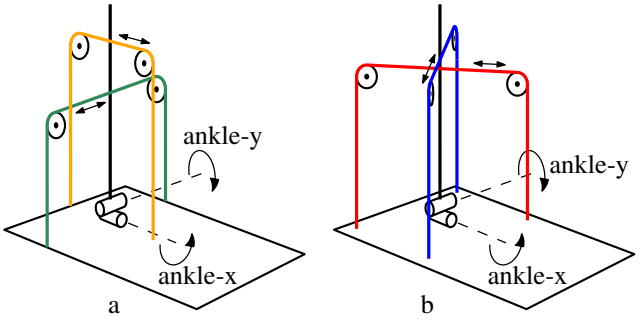


Fig. 6. Ankle actuation concepts. (a) In two separate systems, one motor drives the ankle-x and another the ankle-y joint. (b) In a differential setup the sum of the actuator motions drives the ankle-y and the difference the ankle-x.

Fig. 5 shows our new system. Two Maxon RE25 DC motors are connected to the foot through cables with series-elastic springs. The motors are mounted in a differential set-up: both motors act on both the x-axis and y-axis simultaneously. The cables are guided along 'wings' that are mounted on the sides of the leg, so that the lower leg can flex without the wings touch the upper leg.

4.1 Series elastics

Series elastic actuation can be implemented in different ways. For the new ankle actuation design, the same implementation is used as on the other SEA joints on TUlip: a system consisting of pulleys and cables with springs. It has proved to function well, and routing of the cables gives freedom in the placing of the actuator at a convenient place.

4.2 Differential setup

Conventionally, one would simply use one motor for actuation of the x-axis and the other motor for actuation of the y-axis (Fig. 6a). We chose however to merge the two axes in a so-called *differential setup*, where both motors act on both axes (Fig. 6b). When the motors turn in the same direction, the foot rotates about the y-axis; when they turn in different directions, the foot rotates about the x-axis.

The advantage of the differential setup is that both motors can work together in the joint direction that has the largest demand on the torque. Therefore the required maximum torque per motor is lower and lighter motors can be used for the same performance. A consequence is that the maximum torque in diagonal direction decreases to the torque of a single one of the actuators. Also, a more complex (MIMO) controller has to be designed because the motions of the x-axis and y-axis are coupled.

4.3 Actuator choice

For TUlip the RE line-up of motors by Maxon is used. They have a good performance versus weight ratio and there is good experience with them at the department. We considered two motors: the RE25 (a 20 W motor) and the RE30 (a 60 W motor). Smaller motors are unlikely to be powerful enough and stronger models are unlikely to be needed, and would certainly be too heavy.

Parameter	Name	RE25	RE30	Unit
Rotor inertia	J_{mot}	10.7	33.3	gcm^2
Nominal speed	$\omega_{\text{mot,max}}$	875	842	rad/s
Maximum torque*	$\tau_{\text{mot,max}}$	129	259	mNm
Mass	m_{mot}	130	238	g

*For the RE25 we took half the stall torque as the maximum torque (this is already much more than the allowable continuous torque). For the RE30 we took the torque at $I = 10$ A as maximum torque, as that is the maximum that our motor amplifiers can deliver.

Table 2. Parameters of the Maxon RE25 and Maxon RE30 motors.

The amount of acceleration that a motor can produce is not only dependent on the amount of torque it can generate, but also on the actuator's own inertia (this is the reason that we expressed the actuator demands in acceleration instead of torque in section 3.1). The angular acceleration that can be achieved by a DC motor with gearbox ratio n and gearbox efficiency $\eta = 0.7$ is

$$\ddot{\varphi}_{\text{max}} = \frac{n\,\tau_{\text{mot,max}}\,\eta}{J + n^2 J_{\text{mot}}}. \tag{9}$$

The maximum angular velocity of an actuator is simply

$$\dot{\varphi}_{\text{max}} = \tfrac{1}{n}\omega_{\text{mot,max}}. \tag{10}$$

With these two equations and the motor parameters (table 2), we can plot a point in Fig. 4 and read, for the specific actuator, the maximum achievable walking velocity. In the figure we have plot three lines, representing a single Maxon RE25, a Maxon RE25-pair in differential setup and a single Maxon RE30 motor, all with varying gearbox ratio.

It seems that we can get approximately the same performance for a single RE30 and two RE25s in differential setup. The difference is that, in the case of one RE30 driving ankle-y we need another motor (an RE25) for ankle-x, while in the differential setup, two smaller motors suffice for both axes. Hence, the total system is lighter without giving in on performance.

The ideal transmission ratio of 1:117 should be the total ratio between revolutions of the DC motor axis and the foot. On top of the gearbox, there is also a small pulley-system involved, of which the transfer ratio can be freely chosen (up to some limits). The total ratio is then the product of the gearbox ratio and the pulley system ratio.

5. CONTROL

Control of 'normal' series elastic actuation systems, such as in the hips and knees of TUlip has been studied intensively (Williamson, 1995; Robinson, 2000). Our new ankle actuation system differs from the 'normal' series elastics in two ways:

- The system consists of two coupled series elastics, *i.e.*, it is a multiple-input-multiple-output (MIMO) system,
- The system is nonlinear with respect to ankle position.

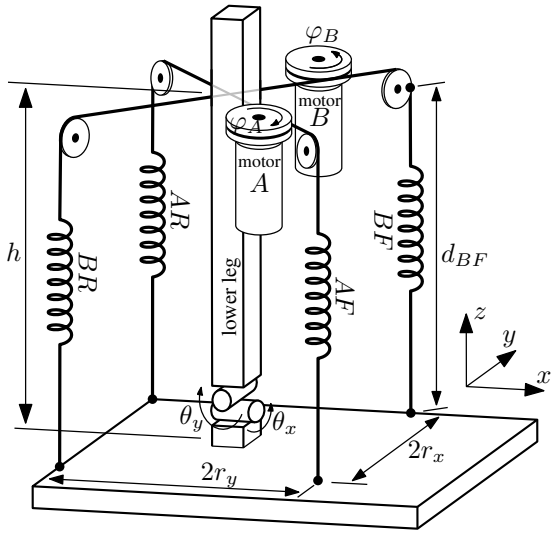


Fig. 7. Sketch of the new ankle mechanism.

5.1 Linear control of the coupled Series Elastic Actuator

Firstly, we look at the system linearized around its 'home' position: $(\theta_x, \theta_y) = 0$. Therefore we use the small angle approximation $\sin\theta_\bullet \approx \theta_\bullet$ and $\cos\theta_\bullet \approx 1$. In that case, the following equations hold:

$$\begin{aligned} J_m\,\ddot{\varphi}_A &= \tau_A + r\,(F_{AF} - F_{AR}),\\ J_m\,\ddot{\varphi}_B &= \tau_B + r\,(F_{BF} - F_{BR}), \end{aligned} \tag{11}$$

and

$$\begin{aligned} \tau_x &= r_x\,((F_{AR} - F_{AF}) - (F_{BR} - F_{BF}))\,,\\ \tau_y &= r_y\,((F_{AR} - F_{AF}) + (F_{BR} - F_{BF}))\,, \end{aligned} \tag{12}$$

where J_m is the apparent inertia of each actuator, $\tau_{A/B}$ is the actuator torque of motor A/B (on the output shaft of the gearbox), $\tau_{x/y}$ is the resulting torque on the x- and y-axis of the foot, r the radius of the pulley on each actuator, $r_{x/y}$ the arm about which the cables exert their force on the foot and $F_{\bullet F}$ and $F_{\bullet R}$ represent the contracting force of the front and rear springs in cables A and B respectively. The elongation of each of the springs, $\Delta l_\bullet$, can be determined as

$$\Delta l_{BF} = -r\varphi_B + (d_{BF} - d_{BF,0}), \tag{13}$$

and similarly for the other springs (with care for the minus-sign in front of r). Here, d_{BF} is the euclidean distance between the attachment point of the spring on the foot and the top pulley, as indicated in Fig. 7 and $d_{BF,0} = h$ is the initial distance (at $\theta_x = \theta_y = 0$). In order to make sure that there is always a positive tension on each of the cables, we use springs having a pre-tension F_{pre}. So the contracting force of the each spring is equal to

$$F_\bullet = k\,\Delta l_\bullet + F_{\text{pre}} \tag{14}$$

where k is the spring constant of each of the springs. In the case of a linear approximation, we have, explicitly

$$\begin{aligned} F_{AF} &= k\,(-r\varphi_A + r_x\theta_x + r_y\theta_y) + F_{\text{pre}},\\ F_{AR} &= k\,(\;\; r\varphi_A - r_x\theta_x - r_y\theta_y) + F_{\text{pre}},\\ F_{BF} &= k\,(-r\varphi_B - r_x\theta_x + r_y\theta_y) + F_{\text{pre}},\\ F_{BR} &= k\,(\;\; r\varphi_B + r_x\theta_x - r_y\theta_y) + F_{\text{pre}}. \end{aligned} \tag{15}$$

By manipulating these equations we find

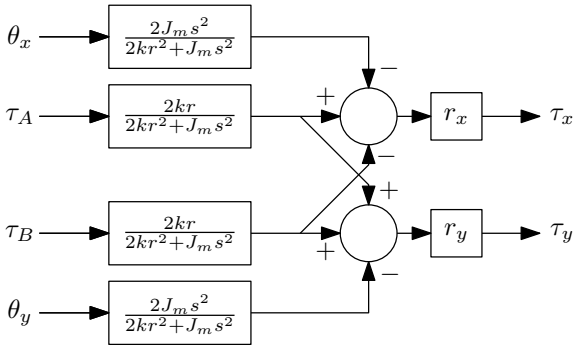


Fig. 8. Linear plant model of the coupled series elastic ankle actuation system.

$$\begin{aligned} T_x &= \frac{2kr_x\,\left(rT_A - rT_b - 2J_m r_x \Theta_x s^2\right)}{2kr^2 + J_m s^2},\\ T_y &= \frac{2kr_y\,\left(rT_a + rT_b - 2J_m r_y \Theta_y s^2\right)}{2kr^2 + J_m s^2}, \end{aligned} \tag{16}$$

where $T_\bullet$ and $\Theta_\bullet$ are the Laplace transforms of $\tau_\bullet$ and $\theta_\bullet$ respectively. If we look at the static case, *i.e.*, $\lim_{s\to 0}$, we find for the input sensitivity S, *i.e.*, the ratio between input torque $\tau_{A/B}$ and output torque $\tau_{x/y}$:

$$\begin{aligned} S_{xA} &= \frac{\partial\tau_x}{\partial\tau_A} = \frac{r_x}{r} & S_{xB} &= \frac{\partial\tau_x}{\partial\tau_B} = -\frac{r_x}{r},\\ S_{yB} &= \frac{\partial\tau_y}{\partial\tau_A} = \frac{r_y}{r} & S_{yB} &= \frac{\partial\tau_y}{\partial\tau_B} = \;\; \frac{r_y}{r}. \end{aligned} \tag{17}$$

The linearized model of the plant is shown in Fig. 8. For controlling the system it is relatively straightforward to invert the transfer functions and take them as a feed-forward term, such as in (Williamson, 1995). Then, by linear feedback, disturbances and model mismatch can be suppressed. However, as we will see in the following subsection, the non-linear effects are so large that it is wise to compensate for them with a decent non-linear feed-forward controller.

5.2 Nonlinearity

In the system we recognize two main sources of non-linearity:

(1) The cable attachment points on the foot describe a sphere under variation of θ_x and θ_y, *i.e.*, the small angle approximation for these variables does not hold exactly,
(2) The rotation axes of the foot are not in the same plane as the cable attachment points, which also gives rise to non-linearities.

The goal of the following subsections is to investigate how bad the non-linearities are and (thus) how much effort should be put in compensating for these non-linearities (especially because we don't want to use too much computational power, if unnecessary).

It should be noted that the 'top' part of the system (the motors with pulleys) does show linear behavior and we consider the springs as having a linear force-elongation relationship. Therefore, (11), (13) and (14) still hold during the non-linear analysis. The other equations of section 5.1 are linear approximations and will be replaced by non-linear equivalents.

Parameter	Name	Value
Arm x-rotation	r_x	5.5 cm
Arm y-rotation	r_y	6.0 cm
Motor pulley radius	r	1.65 cm
Mechanism height	h	22.5 cm
x-axis height	H_x	0.5 cm
y-axis height	H_y	1.6 cm
Spring pretension	F_{pre}	500 N
Spring stiffness	k	50 kN/m
Rotation range x	θ_x	$[-23° \dots 20°]$
Rotation range y	θ_y	$[-32° \dots 35°]$

Table 3. Some important dimensions of the ankle mechanism.

5.3 Nonlinearity — releasing the small-angle approximation

The effect of releasing the small angle approximation can be investigated by replacing (12) by

$$\begin{aligned}\tau_x &= (r_x \cos\theta_x)\,(F_{AR} + F_{BF} - F_{AF} - F_{BR}),\\ \tau_y &= (r_y \cos\theta_y - r_x \sin\theta_x \sin\theta_y)\,(F_{AR}+F_{BR}-F_{AF}-F_{BF})\end{aligned} \tag{18}$$

and replacing $r_x\theta_x$ and $r_y\theta_y$ in (15) by $(r_x \sin\theta_x \cos\theta_y)$ and $(r_y \sin\theta_y)$ respectively. We assume here that the cables still pull exactly vertically (which can be assumed valid if $h \gg r_x\theta_x, r_y\theta_y$). By doing the math we find

$$\begin{aligned}\tau_x &= \frac{r_x}{r}\cos\theta_x(\tau_A - \tau_B),\\ \tau_y &= \frac{r_y}{r}\cos\theta_y(\tau_A + \tau_B) - \frac{r_x}{r}\sin\theta_x \sin\theta_y(\tau_A - \tau_B)\end{aligned} \tag{19}$$

and the input sensitivity S for the static case with nonlinearity, being

$$\begin{aligned}S_{xA} &= \frac{\partial\tau_x}{\partial\tau_A} = \frac{r_x}{r}\cos\theta_x, \qquad S_{xB} = \frac{\partial\tau_x}{\partial\tau_B} = -\frac{r_x}{r}\cos\theta_x,\\ S_{yA} &= \frac{\partial\tau_y}{\partial\tau_A} = \frac{r_y}{r}\cos\theta_y - \frac{r_x}{r}\sin\theta_x\sin\theta_y,\\ S_{yB} &= \frac{\partial\tau_y}{\partial\tau_B} = \frac{r_y}{r}\cos\theta_y + \frac{r_x}{r}\sin\theta_x\sin\theta_y.\end{aligned} \tag{20}$$

Due to this non-linearity, the input sensitivity of the system has become dependent on de foot angle. Note however, that the system is still linear in $\tau_{A/B}$ (*i.e.*, the system can be written as $\tau_x = S_{xA}(\theta_{x/y})\,\tau_A + S_{xB}(\theta_{x/y})\,\tau_B$ and similarly for τ_y). For the dimensions of our system, as stated in table 3, the input sensitivity S_{yA} is shown in Fig. 9. The maximum deviation from the linear case is approximately 32%, which is substantial.

5.4 Nonlinearity — off-plane rotation axes

Fig. 7 shows that the rotation axes of the ankle are not exactly in the plane of the cable attachment points; they are a little higher (also indicated by H_y in Fig. 10). The reason that we designed it like this is that we wanted to foot to be as light as possible, thus make it not too high. The rotation axes need some clearance from the ground however, otherwise the encoders would touch the ground. This gives rise to extra non-linearities, and as we will see in this section, even instability.

In order to be able to visualize everything, we will limit ourselves here to rotation about the y-axis, thus throughout this section we assume that $\theta_x = 0$ and that (by proper control) $\varphi_A = \varphi_B$ (which we will denote as $\varphi_{A/B}$) and $\tau_A = \tau_B$ (ditto). Both springs AF and BF act identically; the same holds for springs AR and BR. We will denote them here as $\bullet F$ and $\bullet R$. Furthermore, again we only consider the static case.

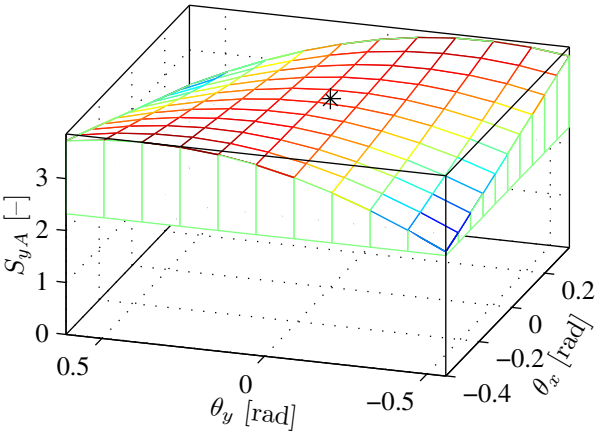


Fig. 9. The input sensitivity $S_{yA} = \frac{\partial\tau_y}{\partial\tau_A}$ as a function of the foot angles θ_x and θ_y, for the non-linearity described in section 5.3. In the fully linear case, the surface would be exactly flat, meaning that the relation between τ_y and τ_A is constant, thus independent on the foot angles. The point $(\theta_x, \theta_y) = (0, 0)$ is indicated with a star.

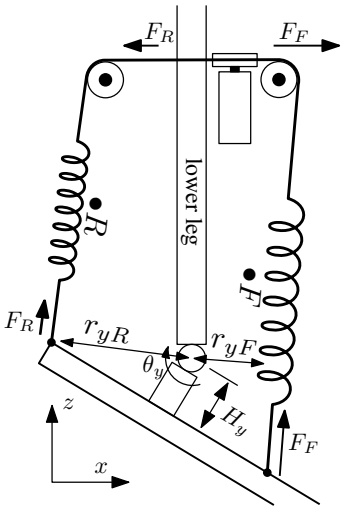


Fig. 10. Example of positioning the foot at an angle θ_y. Because the rotation axis does not lie in the plane of the spring attachment points, the arms r_{yF} and r_{yR} are dependent on θ_y. As a result, in order to get zero torque on the rotation axis, the spring tensions F_R and F_F are not equal, resulting in an acceleration of the motor (unless a counteracting torque $\tau_{A/B}$ is applied).

Assume that the foot is at some angle θ_y and we want to keep it there by making sure that $\tau_y = 0$ (see Fig. 10). From the figure it immediately becomes clear that $F_{\bullet F} \neq F_{\bullet R}$ because the arms for the front and rear springs are different. From (11) it follows that, in order to get no motor acceleration ($J_m\,\ddot{\varphi}_{A/B} = 0$), we need a static torque $\tau_{A/B}$ to compensate for the force mismatch in the springs.

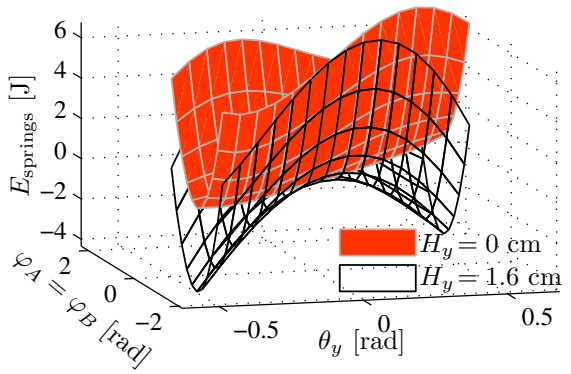


Fig. 11. The total potential energy E_{tot} of the four springs together as a function of θ_y and $\varphi_{A/B}$. for a system where the rotation axis lies in ($H_y = 0$ cm) and outside ($H_y = 1.6$ cm) the plane with the spring attachment points. The other parameters used are the actual parameters of our robot.

Apparently, due to this non-linearity, the system is not linear in torque anymore: for a zero torque τ_y we need a non-zero $\tau_{A/B}$. We will investigate this phenomenon in more detail.

Instead of approaching Δl linearly, we now look at the fully nonlinear function $\Delta l = f(\theta_x, \theta_y, \varphi_A, \varphi_B)$, which in our 2D case reduces to $\Delta l = f(\theta_y, \varphi_{A/B})$. Analytically, this function is quite long, but numerically it can easily be calculated. The potential energy of one pre-tensioned spring can be written as

$$E(\theta_y, \varphi_{A/B}) = \tfrac{1}{2} k\, \Delta l^2 + F_{\text{pre}}\, \Delta l \tag{21}$$

and the total potential energy contained in the four springs together as

$$E_{\text{tot}}(\theta_y, \varphi_{A/B}) = E_{AF} + E_{BF} + E_{AR} + E_{BR}. \tag{22}$$

Fig. 11 shows the total potential energy E_{tot} as a function of its variables for the case where the rotation axis is on ($H_y = 0$ cm) and off ($H_y = 1.6$ cm) the plane of the spring attachment points. We discuss both cases separately, starting with $H_y = 0$ cm. As the system naturally strives towards minimum energy, it will flow towards the valley and stay there; the system has reached a neutral equilibrium (this is actually a way to control the angle θ_y: control the motors to an angle $\varphi_{A/B}$ for which $(\theta_y, \varphi_{A/B})$ is a minimum of the energy function. The system then will converge to the desired foot angle). For $H_y = 1.6$ cm, there is only one (even unstable) equilibrium point: $(\theta_y, \varphi_{A/B}) = (0, 0)$. For any non-zero foot angle, the system is unstable; it will diverge towards one of the end stops. The force with which this occurs is linearly dependent on the pretension in the springs. The partial derivative

$$\frac{\partial E_{\text{tot}}}{\partial \varphi_A} = \tau_A \tag{23}$$

tells us the amount of torque on the motor axis that is 'generated' by the non-linearity. This torque is shown in Fig. 12, as a function of both foot angles. In order to not let the motor axis accelerate by this torque, we can make our controller counteract it with an opposite (feed forward) torque. The torques are substantial (and render the system unstable), up to 2 Nm. Therefore it is important to indeed compensate for it.

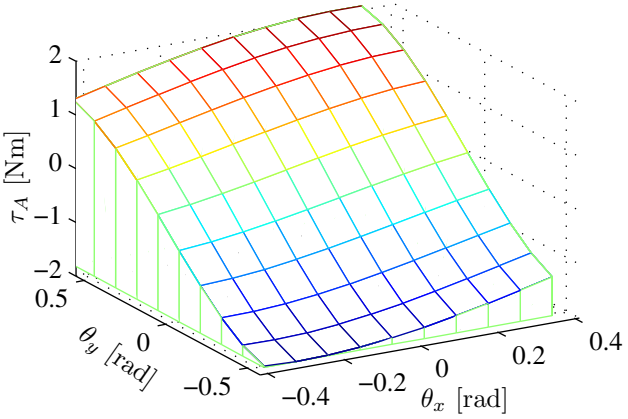


Fig. 12. The torque $\tau_A = \frac{\partial E}{\partial \varphi_A}$ 'generated' by the non-linearity in the system. The torque τ_B has a similar profile but is mirrored around $\theta_x = 0$.

6. CONCLUSIONS

In this paper we discussed the design and control of a new ankle mechanism for the humanoid robot TUlip. It has been shown that the problems of the previous design could be eliminated by using a differential motor setup, resulting in a coupled MIMO system. Calculations were done to determine the best motor and gearbox ratio, where achievable walking speed was chosen as optimization criterion. Some issues on controlling the system were discussed, mainly the non-linearities induced by the system.

We have not yet been able to do intensive testing with the new ankle system, therefore, we cannot tell yet how well the system behaves. Mechanically, we think the new system will easily outperform its predecessor. From a control point of view, the non-linearities can be compensated in software, but we will probably also try to reduce the amount of non-linearity in the system by moving the cable attachment points such that the rotation axes are in the same plane.

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